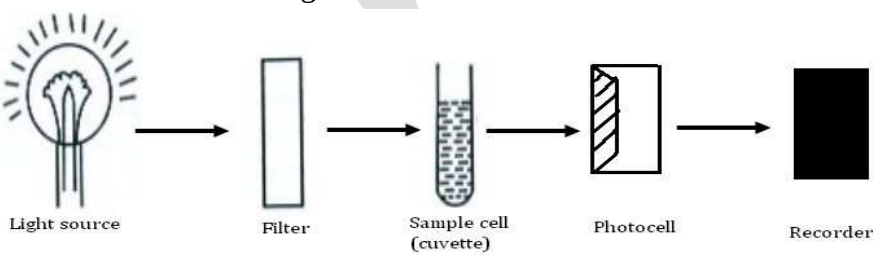
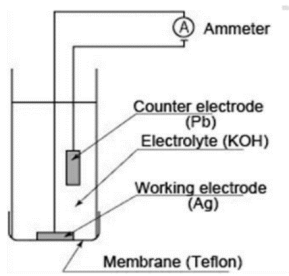




Department of Chemistry
Model Question Paper-1 with solution (2022) scheme

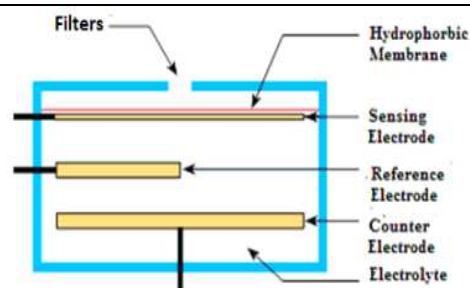
1. a	Explain the working principle of Conductometric sensors (Conductometry), and Optical sensors (colorimetry) Conductometric Sensors It involves the determination of the concentration of analyte based on the measurement of changes occur in electrolyte solution. Conductance is depending on ✓ No. of ions ✓ Mobility of ions Electrode used is conductivity cell. It is made up of two platinum foils with unit cross sectional area and unit distance between them. Conductance of unit volume of the solution is called specific conductance and it is given by $k = \frac{1}{R} \times \frac{l}{a}$ Here l/a is known as cell constant, R is resistance. Optical Sensors: These sensors based on the interaction of electromagnetic radiation with the chemical species. Commonly UV-Visible-Infrared electromagnetic radiations are used. In an optical sensor, the optical signal arises from the interaction of the analyte with an incident radiation. This interaction could results in absorption, emission, scattering and reflection of light. The intensity of the radiation gives the information on the concentration of the analyte. 1. Optical sensors are used to determine the concentration of coloured solution. It is based on the measurement of absorbance of the coloured solution at particular wavelength. It is governed by Beer-Lambertz law. 2. The optical sensors components are light source, filters, photocell and display system. Schematic diagram is as follows  The diagram illustrates the components and flow of an optical sensor system. It starts with a 'Light source' (represented by a sun-like icon), followed by a 'Filter' (a vertical rectangle), then a 'Sample cell (cuvette)' (a test tube containing a liquid), then a 'Photocell' (a rectangle with diagonal lines), and finally a 'Recorder' (a solid black rectangle). Arrows indicate the sequential flow of light through these components.	07 M
b	What are Electrochemical Sensors? Explain its application in the measurement of Dissolved Oxygen (DO) Electrochemical Sensors: These sensors use electrode as transducer component. The main components of electrochemical sensors are working or sensing electrode, electrolyte, counter and reference electrode. Electrolyte is a	

	part of the electrochemical Sensors and role is to transport charge within the sensors, contact all electrode effectively, it is stable under all conditions. Electrochemical Sensors: In electrochemical sensors, two electrodes are used. Anode: Zn, Pb or any other active metal Cathode: Working electrode-Ag Electrolyte: KOH, NaOH or any other inert electrolyte Membrane: Teflon Working: The difference in potential between the anode and the cathode should be at least 0.5V. When electrode is dipped in water to measure DO, anode undergoes oxidation liberating electrons $2Zn \rightarrow 2Zn^{2+} + 4e^{-}$ At cathode, DO undergo reduction. Ag cathode is inert, it only passes electrons to oxygen for reduction. $O_2 + 2H_2O + 4e^{-} \rightarrow 4OH^{-}$ Overall reaction is $2Zn + O_2 + 2H_2O \rightarrow 2Zn(OH)_2$ The current produced by the reduction of oxygen at cathode is proportional to the oxygen in the water sample.	 07 M
2. a	Explain the working principle of Electrochemical sensors, and mention its applications Electrochemical Sensors: These sensors use electrode as transducer component. The main components of electrochemical sensors are working or sensing electrode, electrolyte, counter and reference electrode. Electrolyte is a part of the electrochemical Sensors and role is to transport charge within the sensors, contact all electrode effectively, it is stable under all conditions. Following steps are involved in working of an electrochemical sensor 1. Diffusion of the analyte to the electrode/electrolyte interface (in the liquid form) 2. Adsorption onto the electrode surface 3. Electrochemical reaction with electron transfer 4. Desorption of the product 5. Diffusion of the products away from the reaction zone to the bulk of electrolyte or gas phase. Applications: 1. The oxygen sensors are used to determine dissolved oxygen in boiler water and to monitor dissolved oxygen concentrations in hydrogen fuel cell. 2. Used in security and defense applications like detection of toxic gases. 3. Used in water analysis and environmental monitoring. 4. Used in diagnostic and health care applications. 5. Used in soil parameter analysis and in agricultural applications.	06
b	Describe the application of Electrochemical gas sensors in sensing SO_x and NO_x Electrochemical gas sensors for SO_x and NO_x. Filters: Used to prevent unwanted contaminants, mainly particulate matter Membrane: A gas-permeable membrane is used to regulate the gas flow into the sensors. It allows only	

analyte gas to pass and prevent the leakage of the electrolyte.

Electrodes: two or three electrodes are used on the requirement. Working or sensing, counter and reference electrode.

Electrolyte: Electrolyte should be ionic conductor and chemically stable. Main role is, it transport charge within the sensor, contact all electrodes effectively and solubilise the reactant and product for efficient transport.



Sensors for SO_x:

The sensors contains two or three electrodes

Sensing electrode: Au/Nafion

Electrolyte: 0.5M H₂SO₄

Working:

1. The diffusion of gas analyte through filter, membrane and then finally through electrolyte on to the surface of sensing electrode.
2. Adsorption of analyte gas molecules on the surface of sensing electrode.
3. Oxidation of analyte on the surface of sensing electrode, liberating electrons.
4. Desorption of product from the electrode surface.
5. Diffusion of the products away from the reaction zone to bulk of electrolyte.



Sensors for NO₂:

The sensors contain two or three electrodes.

Sensing electrode: Au, Pt/Nafion.

Electrolyte: 10 M H₂SO₄

Sensors for NO:

The sensors contain two or three electrodes.

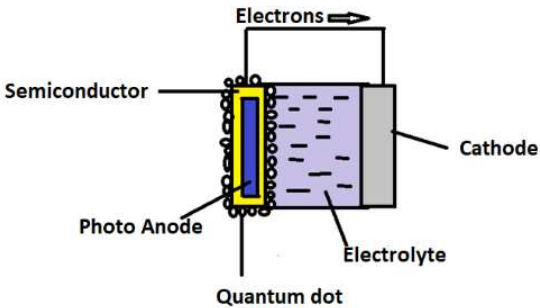
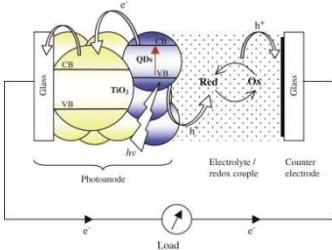
Sensing electrode: Au/NASICON.

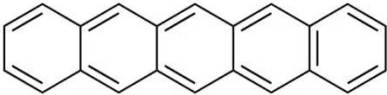
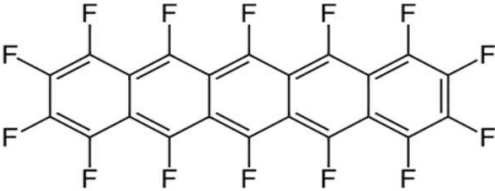
Electrolyte: NaNO₂

Working:

1. The diffusion of gas analyte through filter, membrane and then finally through electrolyte on to the surface of sensing electrode.
2. Adsorption of analyte gas molecules on the surface of sensing electrode.
3. Oxidation of analyte on the surface of sensing electrode, liberating electrons.
4. Desorption of product from the electrode surface.
5. Diffusion of the products away from the reaction zone to bulk of electrolyte.



c	What are Quantum Dot Sensitized Solar Cells (QDSSC's)? Explain the working Principle, Properties and Applications.	
	QUANTUM DOT SENSITIZED SOLAR CELLS (QDSSC) - Quantum dots are considered to be artificial atoms. - Quantum dots (QDs) are semiconductor particles a few nanometres in size, having optical and electronic properties that differ from those of larger particles. - A quantum dot solar cell (QDSC) is a solar cell that uses quantum dots as the absorbing photovoltaic material. - It is used to replace bulky materials such as silicon, or copper indium gallium selenide. - Quantum dots have band gaps that are adjustable through a wide array of energy levels by changing the size of the dots. Construction: Photo Anode: It is conducting glass over which semiconductor is coated (TiO₂). Outer layer of photoanode is coated with quantum (QDs). Electrolyte: Photo anode is contact with redox electrolyte. It is hole conductor. Polysulphide is used as electrolyte. Cathode Electrode: It is used to regenerate electrolyte and complete the circuit. Working: QDs are exposed to sunlight. QDs absorb solar energy, electrons move from valence band to conduction band. These electrons are transferred to semiconductor, leaving behind holes on the surface of QD's - Electrolyte take up the holes from the surface of QD's and get reduced. $S^{2-} + 2h^{+} \rightarrow S$ $S + S_{x-1}^{2-} \rightarrow S_x^{2-} \quad (x = 2 \text{ to } 5)$ - Electrons flows from anode to cathode through external circuit. - At cathode, electrolyte is regenerated taking up electrons from cathode. $S_x^{2-} \rightarrow S + S_{x-1}^{2-}$ $S + 2e^{-} \rightarrow S^{2-}$ Applications of QDSSC: - Used for biological labeling. - Imaging and detection and as efficient fluorescence resonance energy transfer donors. - It is used as light-emitting diodes, photoconductors, photodectors and photovoltaic. - It is used in biomedicine and environment. - It is used in catalysis and other reactions  	07
Module-2		
3. a	Explain the types of organic memory devices by taking p-type and n-type semiconductor materials	07
The p-Type Organic Semiconductor Material “Pentacene”		

	An Organic molecule with π conjugated system and possesses holes as major charge carrier is called p-type semiconductor. Ex: Pentacene These molecules show bistable states when external field is applied i.e. ON and OFF state. It is linearly fused aromatic compound with five benzene rings. It can be obtained in crystal and thin film form. It shows good hole mobility, hence it behaves as a p-type semiconductor.  Pentacene The n-type organic semiconducting material Perfluoropentacene When all the hydrogen atom of pentacene is replaced by Fluorine atoms, it formed Perfluoropentacene. Basically Fluorine is electron withdrawing nature. Hence it converts this molecules into n-type semiconductor.  Perfluoropentacene	
b	What are photoactive and electro active materials and explain their working principle in display system Photoactive and electro active organic materials Organic semiconductors used in electronic and optoelectronic devices are called as electro active and Photoactive materials. Photoactive and electro active organic materials are the semiconductors composed of π-electron systems. Photoactive Process: 1. Absorption and emission of light radiation in the wavelength region from ultraviolet to near infrared. 2. Photo generation of charge carriers (photons of light creates electron-hole pair in the semiconductor) 3. Transport of charge carriers (Charge carriers are particles or holes that freely move within a material and carry an electric charge) Electro active Process Injection of charge carriers from the electrode (The process whereby light is emitted at the junction of N- and P-type semiconductors when an external electric source is applied to drive the electrons and the holes into the junction) Transport of Charge carriers	06
c	What are nanomaterials? Explain any four properties of Polythiophene(P3HT) suitable for optoelectronic devices.	07

Nanomaterials (Silicon Nanocrystals) for Optoelectronic devices

Any substance in which at least one dimensions is less than 100nm is called nanomaterials.

Organic materials for Optoelectronic devices [Light absorbing materials –Polythiophene (P3HT)]

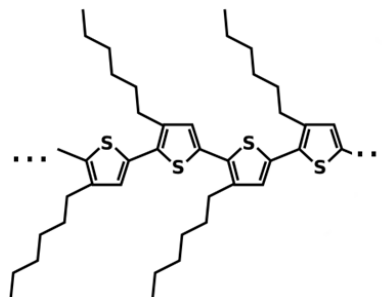
Polythiophene are an important class of conjugated polymers, environmentally and thermally stable material. Chemical structure of P3HT Poly (3-hexylthiophene) is a polymer with chemical formula $(C_{10}H_{14}S)_n$. It is a Polythiophene with a short alkyl group on each repeat unit.

Highly ordered (P3HT) are composed of closely packed, p-p stacked (p-p distance of 0.33nm) fully extended chains which are oriented perpendicular to the substrate.

Structure of P3HT

Properties:

1. P3HT is a semiconducting polymer with high stability and exhibits conductivity due to holes therefore considered as p-type semiconductor.
2. Poly-3-hexylthiophene (P3HT) has great capability as light-absorbing materials in organic electronic devices.
3. P3HT has a crystalline structure and good charge-transport properties required for Optoelectronics.
4. P3HT has a direct-allowed optical transition with a fundamental energy gap of 2.14eV.
5. Fundamental band gap of P3HT is 490nm visible region, corresponding to $\pi \rightarrow \pi^*$ transition, giving electron-hole pair.
6. P3HT indicates that an increase in the conductivity is associated with an increase in the degree of crystallinity.



OR

4. What are Memory Devices? Explain the Classification of electronic memory devices with examples

06

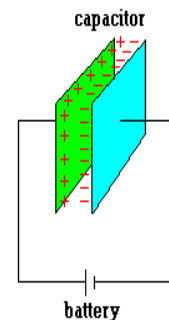
Classification of Electrical (electronic) Memory Devices : Electronic memory devices can be divided into 4 types depending the type of material it is made of.

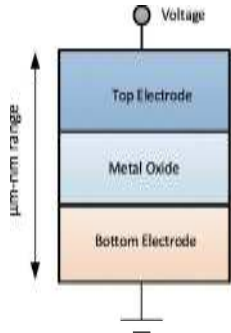
1. Transistor-Type Electronic Memory Devices
2. Capacitor-Type Electronic Memory Devices
3. Resistor-Type Electronic Memory Devices
4. Charge Transfer Type Electronic Memory Devices

Transistor-Type Electronic Memory

Transistors are made from silicon, a semiconductor. It is converted to p-type and n-type semiconductor by doping trivalent and pentavalent impurities. Transistors are made using p-type and n type semiconductor. A transistor is a miniature electronic component that can work either as an amplifier or a switch. A computer memory chip consists of billions of transistors; each transistor is working as a switch, which can be switched ON or OFF .Each transistor can be in two different states and store two different numbers, ZERO and ONE. Since chip is made of billions of such transistors and can store billions of Zeros and ones ,and almost every number and letter can be stored.

Capacitor-Type Electronic Memory



	A capacitor consists of two metal plates which are capable of storing an electric charge. It is used to store data. It is like a battery that holds data based on energy. If the capacitor is charged, it holds the binary numeral, “1” and holds “0” when the cell is discharged. If the parallel plates of a capacitor are separated by dielectric layer, charges dissipate slowly and memory would be volatile. On the other hand, if the medium between the electrodes is ferroelectric in nature, can maintain permanent electric polarization that can be repeatedly switched between two stable states (bistable) by an external electric field. Thus, memory based on ferroelectric capacitors (FeRAM) is non-volatile memory. Resistor-Type Electronic Memory Memory devices containing switchable resistive materials are classified as resistor-type memory, or resistive and random access memory (RRAM). Resistor-type electronic memory usually has a simple structure, having a metal-insulator-metal structure generally referred to as MIM structure. The structure comprises of an insulating layer (I) sandwiched between the two metal (M) electrodes and supported on a substrate (glass, silicon wafer, plastic or metal foil). Initially, the device is under high resistance state or “OFF” and logically “0” state, when resistance changed or under external applied field changes to lower resistance state or “ON” logical value “1”. Charge Transfer Effects Type Electronic Memory A charge transfer (CT) complex is defined as an electron donor– acceptor (D–A) complex, characterized by an electronic transition to an excited state in which a partial transfer of charge occurs from the donor moiety to the acceptor moiety. The conductivity of a CT complex is dependent on the ionic binding between the D–A components. 	
b	Mention any four properties and applications of LC-displays Properties 1. Liquid crystal can flow like a liquid, due to loss of positional order 2. These are elongated and have some degree of rigidity 3. They have less orientational order 4. Transition from crystalline solids to liquid crystals caused by a change of temperature. Applications of liquid crystals: 1. Liquid crystals are used in watches, calculators, mobile telephones, laptops, computers etc. 2. These are used in blood pressure instruments 3. These are used in potentiometer, conductometer, Colorimeter etc. 4. digital thermometers and TV Channel indicators	07
c	Mention any four properties and applications of QLED Properties: 1. Accurate and vibrant colors: QLEDs are capable of producing highly accurate and vibrant colors due to their use of quantum dots, which emit light of a specific colour when they are excited by a light source or an electrical current. 2. Energy-efficient: QLEDs are more energy-efficient than traditional LCD displays because they do not require as much backlighting. 3. High contrast: QLED displays have high contrast ratios, which means that the difference between the darkest and brightest areas of the display is greater, resulting in more detailed and lifelike images.	07

	4. Long lifespan: QLEDs have a longer lifespan than traditional LCD displays because they do not suffer from the same issue soft back light burn out colour fading overtime. Applications 1. Flat-panel TV screen 2. Digital cameras 3. Mobile phones 4. laptops	
Module-3		
5. a	Define metallic corrosion? Describe the electrochemical theory of corrosion taking iron as an example. Definition: “The spontaneous deterioration and consequent loss of a metal/alloy due to chemical/electrochemical attack by the environment”. Electrochemical theory of corrosion: According to this theory, 1. When a metal is in contact with moist air or when dissimilar metals are contact with each other in a solution, large numbers of galvanic cells are formed with the existence of anodic and cathodic area on the metal. 2. In this corrosion, oxidation of the metal and reduction of species present in solution takes place. 3. The anodic part of the metal undergo oxidation and suffers from corrosion, cathodic part undergo reduction and protected from corrosion. 4. The electrons are transferred through the metal from anode to cathode. 5. During corrosion the following reactions takes place. At anode (oxidation reaction) $M \rightarrow M^{n+} + ne^{-}$ The reaction at cathode (reduction reaction) depends on the nature of the environment: If the medium is acidic, In the presence of dissolved oxygen: $2H^{+} + \frac{1}{2}O_2 + 2e^{-} \rightarrow H_2O$ In the absence of dissolve do xygen: $2H^{+} + 2e^{-} \rightarrow H_2$ If the medium is alkaline/neutral, In the presence of dissolved oxygen: $H_2O + \frac{1}{2}O_2 + 2e^{-} \rightarrow 2OH^{-}$ In the absence of dissolved oxygen: $2H_2O + 2e^{-} \rightarrow 2OH^{-} + H_2 \uparrow$ Example: Rusting of an Iron in the presence of moist air Anodic Reaction: $Fe \rightarrow Fe^{2+} + 2e^{-}$ Cathodic reaction: $H_2O + \frac{1}{2}O_2 + 2e^{-} \rightarrow 2OH^{-}$ Products of anode & cathode: $Fe^{2+} + 2OH^{-} \rightarrow Fe(OH)_2$ In the presence of oxygen: $2Fe(OH)_2 + \frac{1}{2}O_2 \rightarrow Fe_2O_3 \cdot 2H_2O$ (Rust)	07
b	Explain: (i) Differential metal corrosion & (ii) Water-line corrosion Differential metal corrosion (Galvanic corrosion) 1) When two dissimilar metals are in direct contact with one another and exposed to a corrosive medium, the metal with lower electrode potential becomes anode and suffers from corrosion,	06

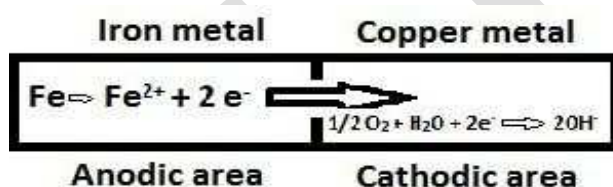
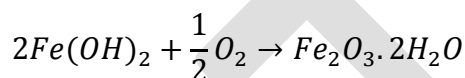
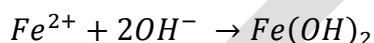
whereas the metal with higher electrode potential becomes cathode and protected from corrosion. This type of corrosion is known as differential metal corrosion or galvanic corrosion.

- 2) The rate of corrosion depends mainly on the difference in the position of the two metals in galvanic series.
- 3) Higher the difference, faster is the rate of corrosion.

Example: Electrode potential of iron (-0.44 V) is less than that of copper (0.34 V). Therefore when iron is in contact with copper, iron becomes anodic and undergoes corrosion, whereas copper becomes cathodic and protected from the corrosion.

At Anode: $Fe \rightarrow Fe^{2+} + 2e^{-}$

At Cathode : $H_2O + \frac{1}{2}O_2 + 2e^{-} \rightarrow 2OH^{-}$



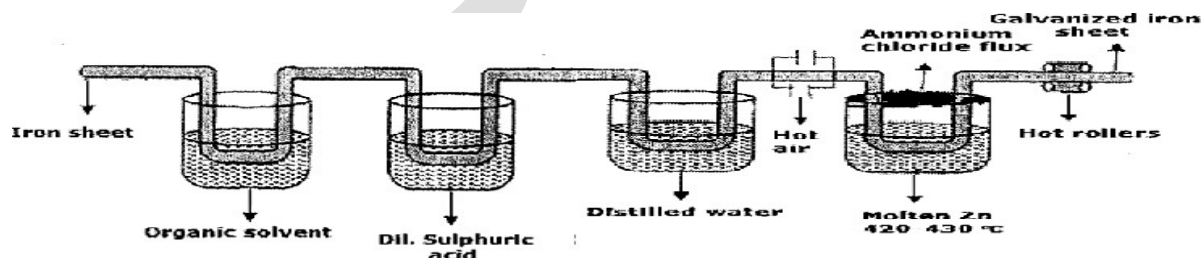
c **Describe galvanizing and mention its applications.**

07

Galvanization

Coating of zinc on iron or steel by hot dipping process is called galvanizing. (M.P of Zn=419 °C) It is carried out as follows,

1. The iron sheet is subjected to pretreatment by degreasing with organicsolvent.
2. The sheet is passed through dilute H_2SO_4 to remove anyrust.
3. Washed with distilled water and dried by supplying hotair.
4. Then iron sheet is dipped in molten bath of Zn at 419°C and surface is covered with NH_4Cl flux which prevents the oxidation of Zn metal.
5. The excess of coated metal is removed by passing through a pair of hot rollers and cooled gradually.



OR

6. **Explain: i) corrosion control by Anodization & ii) Sacrificial anodicmethod.**

06

Anodizing

Anodizing is the process of oxidation of outer layer of metal to its metal oxide by electrolysis. Oxide layer formed over the metal itself acts as protective layer.

Anode : Aluminium

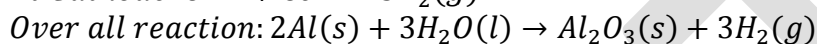
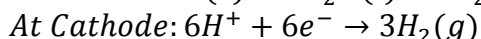
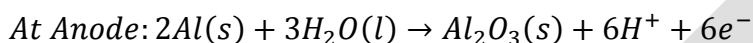
Cathode : Steel or Copper

Electrolyte : 5-10% chromic acid

Temperature : 30-40 °C

Applied potential : 0-40V

Aluminum is made as anode and copper or steel is used as cathode. The electrodes are immersed in an electrolyte consisting of 5-10% chromic acid. The temperature is maintained at 30-40 °C. Potential of 0-40 V is applied which oxidizes outer layer of Al to Al_2O_3 and that get deposited over the metal. The reactions involved are as follows

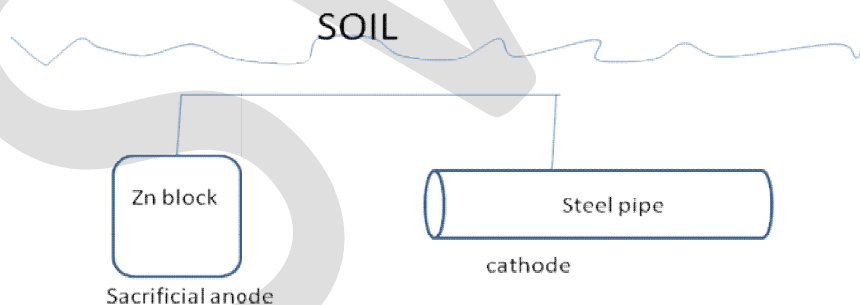


The anodized coatings are thicker than the natural oxide film and possess improved corrosion resistance as well as resistance to mechanical injury.

Sacrificial anodic protection:

1. The metal to be protected is connected to a more anodic metal using a metallic wire.
2. The more active metal gets corroded while the parent structure is protected from corrosion.
3. The more active metal so employed is called sacrificial anode.
4. The sacrificial anodes to be replaced by fresh ones as and when it is required.
5. Commonly used sacrificial anodes are: Mg, Zn, Al etc.

Example: steel pipe is protected by connecting it to a block of Zn. In such cases steel acts as a cathode and is unaffected or protected from corrosion and Zn act as anode and undergoes sacrificial corrosion.



b Explain the construction and working of Calomel electrode

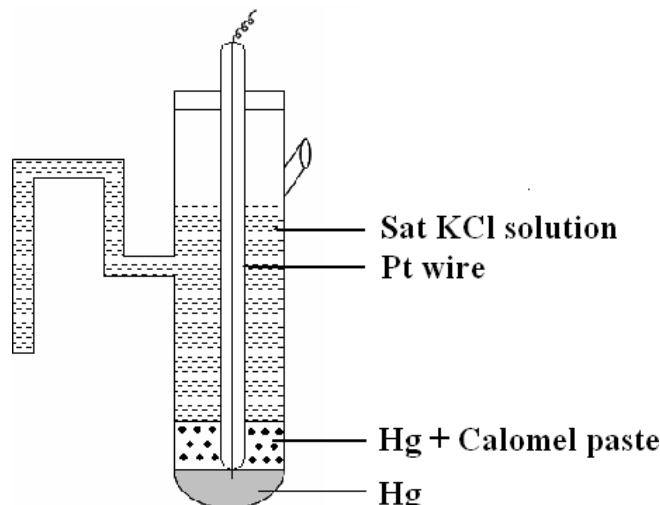
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Calomel Electrode:

Construction:

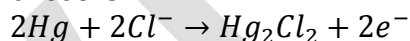
- 1) Calomel electrode consists of long glass tube with two side tubes.
- 2) One at the top to fill saturated KCl solution and the other side tube is connected to the salt bridge. Mercury is placed at the bottom which is covered with a layer of Hg and Hg_2Cl_2 (calomel) paste.
- 3) The remaining portion is filled with saturated KCl solution.
- 4) A platinum wire is dipped into the mercury and is used to provide external electrical contact.

The calomel electrode is represented as, $\text{Cl}^-/\text{Hg}_2\text{Cl}_2/\text{Hg}$

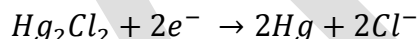


Working:

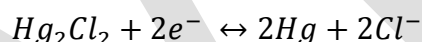
Calomel electrode behaves as anode or cathode depending upon the nature of another electrode. The half-cell reaction when it acts as an anode is



The half-cell reaction when it acts as a cathode is



The electrode reaction is,



Applying Nernst equation,

$$E = E^0 + \frac{0.0591}{n} \log \frac{[\text{Reactants}]}{[\text{Products}]}$$

$$E = E^0 + \frac{0.0591}{n} \log \frac{[\text{Hg}_2\text{Cl}_2]}{[\text{Hg}]^2 [\text{Cl}^-]^2}$$

Since $[\text{Hg}_2\text{Cl}_2] = [\text{Hg}]^2 = 1$, the above equation becomes

$$E = E^0 + \frac{0.0591}{2} \log \frac{1}{[\text{Cl}^-]}$$

The calomel electrode potential is depending on the concentration of chloride ions in KCl. If the Concentration of chloride ions increases, the potential decreases and vice versa. (The potential of calomel is inversely proportional to the concentration of chloride ions)

The potential of calomel electrode is measured with respect to SHE and it depends on concentration of KCl solution used.

Conc. KCl	0.1N	1N	Saturated KCl
Potential (V)	0.334V	0.281V	0.242V

C

What is CPR? A thick brass sheet of area 400 inch² is exposed to moist air. After 2 years of period, it was found to experience a weight loss 375 g due to corrosion. If the density of brass is

06

8.73 g/cm³. Calculate CPR in mpy and mmpy.

Corrosion Penetration Rate (CPR)

The corrosion penetration rate (CPR) is defined as “the speed at which any metal in a specific environment deteriorates due to chemical reactions when it is exposed to a corrosive environment” or the amount of corrosion loss per unit time in thickness or the speed at which corrosion spreads to the inner portions of a material.

To calculate CPR in mpy

	Given	CPR in mpy
K		534
W (wt loss)	375 g	375 x1000 mg
P	8.7g/cm ³	8.7g/cm ³
A	400 inch ²	400 inch ²
T	2.5 years	2.5x365x24 hrs

$$\text{CPR} = \frac{k \times W}{D \times A \times T}$$

$$\text{CPR} = \frac{534 \times 375}{8.73 \times 400 \times 2.5 \times 365 \times 24}$$

$$\text{CPR} = 2.61 \text{ mpy}$$

To calculate CPR in mmpy

	Given	CPR in mmpy
K		87.6
W (wt loss)	375 g	375 x1000 mg
ρ	8.7g/cm ³	8.7g/cm ³
A	400 inch ²	400 x 6.45 cm ²
t	2.5 year	2.5x365x24 hrs

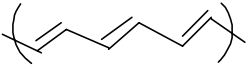
$$1 \text{ inch}^2 = 6.45 \text{ cm}^2$$

$$1 \text{ cm}^2 = 0.155 \text{ inch}^2$$

$$\text{CPR} = \frac{87.6 \times 375}{D \times A \times T}$$

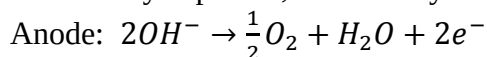
$$\text{CPR} = \frac{87.6 \times 375}{8.73 \times 400 \times 6.45 \times 2.5 \times 365 \times 24}$$

$$\text{CPR} = 3.027 \times 10^{-3} \text{ mmpy}$$

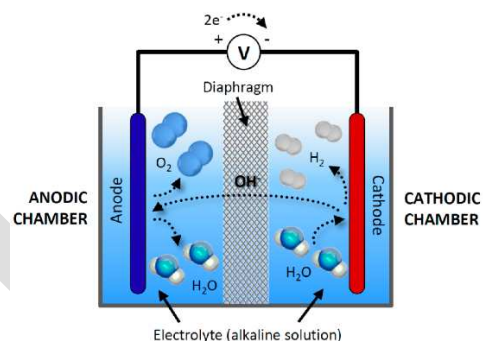
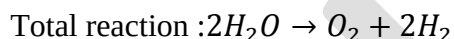
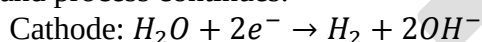
7. a	A polydisperse sample of polystyrene is prepared by mixing three monodisperse samples in the following proportions. 1g of 10000 molecular weight, 2g of 50000 molecular weight and 2g of 100000 molecular weight. Determine number average and weight average molecular weight. Find the index of polydispersity. <u>Number average molecular mass:</u> $\overline{M}_n = \frac{n_1 M_1 + n_2 M_2 + n_3 M_3}{n_1 + n_2 + n_3}$ $\overline{M}_n = \frac{\sum NiMi}{\sum Ni}$ $= \frac{N_1 M_1 + N_2 M_2 + N_3 M_3}{N_1 + N_2 + N_3 \dots}$ $= \frac{1 \times 10000 + 2 \times 50000 + 2 \times 100000}{1 + 2 + 2}$ $= 62000 \text{ g/mol}$ $\overline{M}_w = \frac{N_1 M_1^2 + N_2 M_2^2 + N_3 M_3^2}{N_1 M_1 + N_2 M_2 + N_3 M_3}$ $\overline{M}_w = \frac{1 \times (10^4)^2 + 2 \times 5(10^4)^2 + 2 \times (10^5)^2}{1 \times 10000 + 2 \times 50000 + 2 \times 100000} = 670967.74 \text{ g/mol}$ Poly dispersity index, $\text{PDI} = \frac{\overline{M}_w}{\overline{M}_n} = \frac{670967.74}{62000} = 10.82$	07
b	Explain the synthesis of Polyacetylene and mention its applications From Acetylene: In this method polyacetylene is synthesized by using monomer gaseous acetylene in the presence of Ziegler-Natta catalyst such as Titanium isopropoxide ($\text{Ti}(\text{OPr})_4$) and triethyl aluminium ($\text{Al}(\text{C}_2\text{H}_5)_3$). $\text{CH}=\text{CH}$ Acetylene $\xrightarrow{\text{Ti}(\text{OPr})_4/\text{Al}(\text{C}_2\text{H}_5)_3}$  polyacetylene Commercial Applications of Polyacetylene: - Used in the manufacture of chemical sensors, corrosion inhibitors. - Used in compact electronic devices such as polymer based transistors, LEDs etc.	07
c	Explain the generation of hydrogen by Alkaline water	06

Alkaline water electrolysis

1. It consists of two electrodes i.e. anode and cathode.
2. Both electrodes are made up of Ni based metal, because it is more stable during the oxygen evolution.
3. These electrodes are immersed in KOH solution (25-35%).
4. Both electrodes are separated by porous diaphragm prevent gases crossover and allows only hydroxide ions.
5. Cell voltage is 1.3 – 2 V.
6. When electricity is passed, at anode hydroxide ions lose electrons and forms water molecules.



7. At cathode, water molecules accept electrons and liberate hydrogen gas and forms hydroxide ions.
8. These hydroxide ions move from cathode to anode through diaphragm and process continues.



Advantages

1. Low cost and well established technology
2. Long Stability and stacks in the megawatt range
3. Non-noble catalyst

OR

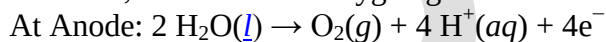
8.
a

Describe the hydrogen production by photo catalytic water splitting method.

Generation of hydrogen by electrolysis of water

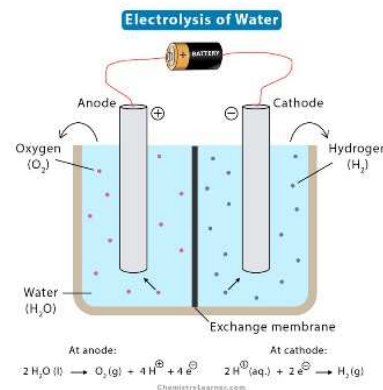
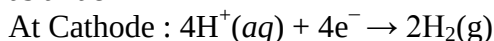
It is the process of splitting of water into oxygen and hydrogen gas by electrolysis.

1. It consists of two electrodes i.e. anode and cathode.
2. Both are separated by membranes.
3. When electricity is passed, oxidation takes place at anode, it gives H^+ ions and electron, also liberates Oxygen gas.



4The H^+ ions move into cathodic compartment through membranes and electrons move from anode to cathode through external circuit.

At cathode the H^+ ions accepts electrons and forms H_2 gas. This liberated hydrogen gas is used as a fuel



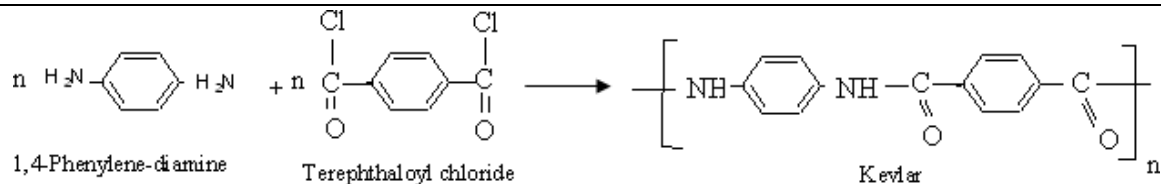
07

b

Preparation, properties, and commercial applications of Kevlar.

Kevlar: Kevlar is synthesized in solution of N-methyl-pyrrolidone & calcium chloride from the monomers 1, 4-phenylene-diamine (para-phenylenediamine) & terephthaloyl chloride through a condensation reaction with liberation of HCl as a byproduct.

07



Properties:

1. It has very light weight.
2. It has high tensile strength and stiffness.
3. It has very good corrosion resistance.

Applications of Kevlar:

1. It is used to make light weight boat hulls
2. aircraft fuselage panels, pressure vassals, high performance race car, bullet proof vests,
3. Used in puncture resistance bicycle tyres etc.

c

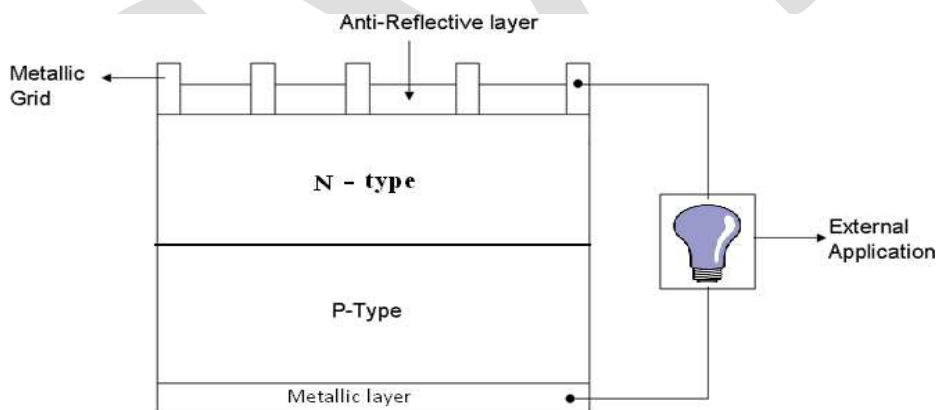
Explain the construction and working of photovoltaic cells.

06

The device, which converts solar energy into electrical energy, is called photovoltaic cell and the phenomenon is called photovoltaic effect.

Construction, working and applications of PV Cell:

Principle: The principle involved is the ejection of electrons from metal surface by striking with photons of solar radiation.



1. The device consists of p-n junction diode coated with anti reflective layer (TiO_2).
2. Two electrical contacts are provided, one in the form of metallic grid at the top of the junction and the other is a layer silver on the bottom of the cell.
3. The antireflective layer coated in between the metallic grids allows light to fall on the semiconductor.

Working:

The photons of solar radiations enters n-type semiconductor breaks barrier potential and moves to p-type semiconductor where photons knocks the electrons in p type to form electron hole pair. The free

	electrons so formed will travels through the circuit from n-type and recombines with holes again in p-region. The movement of electrons from n type to p type generates electric current. The electrical energy produced from the solar cell is used for various applications.	
Module-5		
9. a	Mention the sources of e-waste and explain the need for e-waste <u>Sources of electronic waste (e-waste):</u> The main sources of electronic waste (e-waste) include: 1. Consumer electronics such as smart phones, laptops, televisions, and household appliances. 2. Office equipment such as computers, printers, copiers, and fax machines. 3. Medical equipment such as X-ray machines, monitors, and diagnostic equipment. 4. Electronic toys and games. 5. Military and government surplus equipment. 6. Obsolete technology such as outdated computer equipment and VCRs. 7. Discarded or broken electronic devices. 8. Colleges often use electronic equipment such as projectors, computer equipment, and audio-visual equipment in classrooms and laboratories. <u>Need of E- Waste Management:</u> E-waste management is necessary because it helps to address several environmental, health, and resource-related concerns, including: Protecting the environment: E-waste contains toxic substances, such as lead, mercury, and cadmium that can have harmful effects on the environment and human health if not properly managed. Conserving resources: E-waste contains valuable resources, such as metals, that can be recovered and reused through proper recycling. Reducing greenhouse gas emissions: The production of new electronic products releases greenhouse gases, such as carbon dioxide, into the atmosphere. Proper recycling and disposal of e-waste can reduce the environmental impact of electronic products. Reducing land filling: land filling of electronic waste can result in the release of toxic materials into the environment and contribute to soil and water pollution. Protecting public health: Improper handling and disposal of e-waste can expose workers and the general public to hazardous materials and cause serious health problems. E-waste can be toxic, is not biodegradable and accumulates in the environment, in the soil, air, water and living things. Causes air pollution, Soil pollution and water pollution. Electronic Recycling Promotes Soil Fertility and Maintain Nutrient. E-waste management programs aim to promote responsible recycling and disposal of electronic waste and minimize the release of hazardous Materials into the environment. This helps to protect the environment, conserve resources, and promote public health and safety.	07
b	Explain the recycling of e-waste <u>E- Waste recycling:</u> The process of e-waste recycling typically involves the following steps: 1. Collection and transportation: E-waste is collected from various sources such as households, businesses, and recycling facilities. It is then transported to a recycling plant for processing.	07

	2. Sorting and dismantling: E-waste is sorted into different categories based on the type of material and the manufacturer. The recyclers then dismantle the devices to separate the valuable materials from the hazardous components. 3. Shredding: The e-waste is shredded into smaller pieces to make it easier to separate the different materials. The shredded pieces are then sorted into different categories based on their composition. 4. Separation: The valuable materials, such as metals, plastics, and glass, are separated from the other components through a series of physical and chemical processes. 5. Processing: The separated materials are processed to remove any impurities and contaminants, and to prepare them for reuse. For example, metals are smelted to produce pure metal alloys, while plastics are melted and molded into new products. 6. Disposal of hazardous waste: The hazardous components of e-waste, such as batteries and LCDs, are properly disposed of to prevent pollution and health hazards.	
c	Extraction of gold from e-waste (Explain the Principle and experimental procedure) Principle: The principle behind the extraction of gold from e-waste is based on the fact that gold is a relatively non-reactive metal, which allows it to be recovered from complex electronic waste matrices through a series of chemical and physical processes. Experimental procedure: 1. Collection and segregation of e-waste: The first step involves collecting and segregating the e-waste into different categories, such as computer motherboards, cell phones, and other electronic devices. 2. Physical separation: The e-waste is physically separated into different components, such as plastics, metals, and glass. 3. Leaching: The metals, including gold, are leached from the e-waste using a suitable reagent, such as aqua regia (a mixture of hydrochloric acid and nitric acid), to dissolve the gold. $\text{Au} + \text{HNO}_3 + 4\text{HCl} \rightarrow \text{HAuCl}_4 + \text{NO} + 2\text{H}_2\text{O}$ 4. Precipitation: The dissolved gold is then precipitated out of the solution through the addition of a suitable reducing agent, such as sodium metabisulfite. $2\text{H}[\text{AuCl}_4] + 3\text{Na}_2\text{S}_2\text{O}_5 + 3\text{H}_2\text{O} \rightarrow 2\text{Au} + 3\text{SO}_2 + 3\text{Na}_2\text{SO}_4 + 8\text{HCl}$ 5. Purification: The precipitated gold is then purified through processes such as ion exchange, electro-winning, or distillation, to remove impurities and improve its quality. 6. Recovery: The purified gold is then recovered for reuse.	06
OR		
10 .a	Explain the ill effects of toxic materials used in manufacturing electrical and electronic products Health hazardous due to exposure of e waste: Exposure to electronic waste (e-waste) can have serious health consequences, including: Poisoning: E-waste can contain toxic substances, such as lead, cadmium, and mercury that can cause poisoning if they enter the body. Respiratory problems: Exposure to dust and fumes generated during the dismantling and disposal of e-waste can cause respiratory problems, such as asthma and bronchitis. Neurological effects: Exposure to toxic substances in e-waste, such as lead and mercury, can cause neurological effects, including memory loss,	07

	tremors and coordination problems. Reproductive problems: Exposure to toxic substances in e-waste, such as cadmium, can cause reproductive problems, including infertility and birth defects. Cancer: Exposure to carcinogenic substances, such as dioxins and polychlorinated biphenyls (PCBs), found in e-waste, can increase the risk of cancer It is important to effectively manage e-waste to minimize the exposure of workers and the public to hazardous materials and to prevent serious health problems.	
b	Explain the pyrometallurgical and direct recycling methods. <u>Pyrometallurgical methods E-waste recycling:</u> E-waste pyrometallurgical methods refer to the process of extracting valuable metals and other materials from electronic waste using high temperatures. These methods include: 1. Smelting: The e-waste is melted in a furnace and then separated into individual metals and other materials. 2. Refining: The metals from the smelted e-waste are further processed to remove impurities and improve their quality. 3. Incineration: Electronic waste is burned at high temperatures to reduce its volume and recover metals. Pyrometallurgical methods are effective at recovering valuable metals from e-waste, but they also generate hazardous byproducts and require significant energy inputs. Additionally, these methods can pose a risk to the environment and human health if not properly regulated and monitored. <u>Direct recycling of e waste:</u> Direct recycling of e-waste refers to the process of recovering valuable materials from electronic waste without the need for intermediate processing steps. This can include processes such as shredding, granulating, and sorting, which are used to separate the different components of e-waste, such as metals, plastics, and glass. The separated materials are then processed to extract the valuable components and prepare them for reuse.	06
c	Write a brief note on role of stakeholders for example; producers, Consumers, recyclers, and statutory bodies. <u>Role of stakeholders - producers, consumers, recyclers, and statutory bodies.</u> In the management of electronic waste (e-waste), the following stakeholders play important unique role in the management of e-waste: 1. Producers - are responsible for designing and producing electronic products and may also be involved in the collection and recycling of e-waste. 2. Consumers - play a crucial role in the responsible disposal of e-waste and making informed choices about the purchase of electronic products. 3. Recyclers - are responsible for safely and responsibly managing e-waste, including the collection, dismantling, and recycling of electronic waste. 4. Statutory bodies - such as governments, are responsible for creating and enforcing regulations and policies to manage e-waste and promoting public awareness and education about e-waste management.	07

SVT

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